

TrES-3b

R-2792

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110721 · created 2026-09-04T01:19:05+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455763.78251 +/- 0.00095 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.116 +/- 0.056
Transit depth (Rp/Rs)^2	1.35 +/- 1.29 [%]
Orbital Inclination (inc)	77.08 +/- 2.9
Airmass coefficient 1 (a1)	2.23 +/- 1.42
Airmass coefficient 2 (a2)	-0.21 +/- 1.08
Scatter in the residuals of the lightcurve fit is	68.7 %
Best Comparison Star	#3 - [105, 456]
Optimal Aperture	0.85
Optimal Annulus	4.8
Transit Duration (day)	0.008 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.116 $\pm$ 0.056 vs 0.16309 (deviation 0.66 σ)
Duration fit vs published	0.008 d vs 0.0567 d (deviation 2.25 σ)
Mid-time O-C	-3.1 min
Depth SNR	1.6
Residual scatter	68.7 %

Light curve

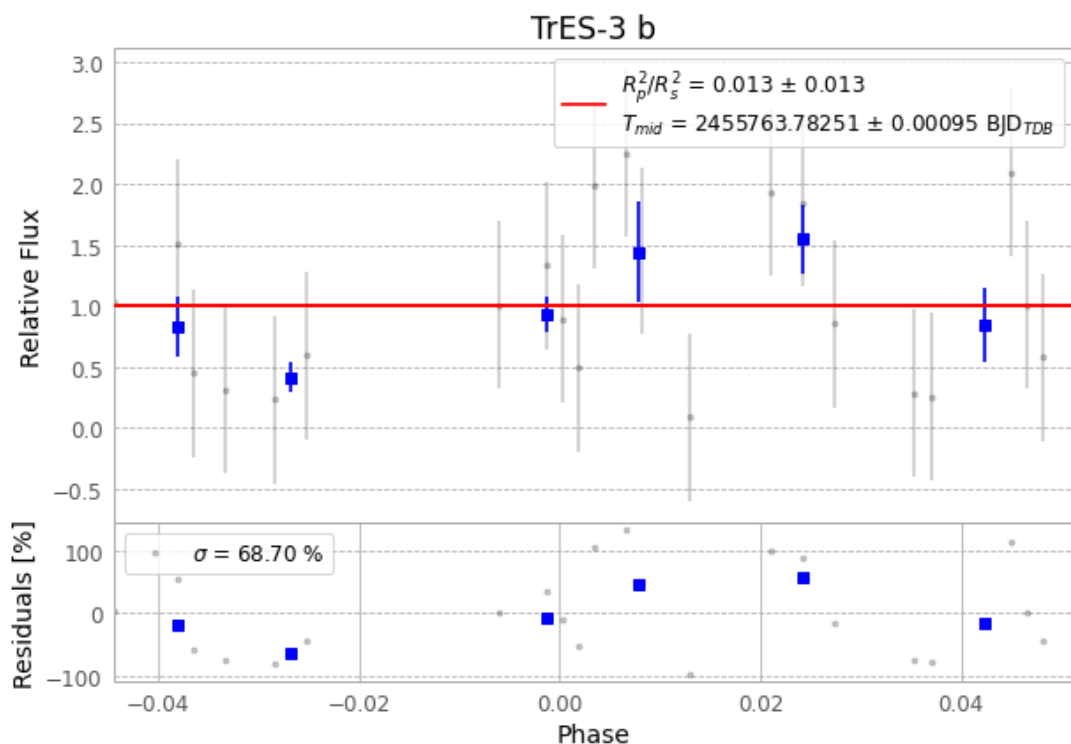


Figure 1 — FinalLightCurve_TrES-3 b_2011-07-20.png (SHA-256 e4b5b08b4602321b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [419, 432], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 88f755142e79e2ff...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

62 FITS frames (41 MB), TRES-3110721051512.FITS → TRES-3110721082712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110721.log (12db773d33a8...), FinalLightCurve_TrES-3 b_2011-07-20.png (e4b5b08b4602...), FinalLightCurve_TrES-3 b_2011-07-20.csv (7ca7fdd42602...)

Compute resources

Wall time 212.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 01:19Z.